INDIAN INSTITUTE OF INFORMATION TECHNOLOGY VADODARA

CS545 – SOFTWARE ENGINEERING



PROJECT REPORT

Hotel Management System

Master of Computer Applications (MCA)

Second Year

Group No. 24

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Certificate

This is to certify that the following students of the Master of Computer Applications (MCA) programme at the

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The report documents the Software Requirements Specification, software process model analysis, modular

design, functional test-case design, and supporting architecture and workflow diagrams for the Hotel

Management System. This certificate section is provided as a template; faculty approval and signature must be completed by the authorized faculty member.

Date: 29/10/2026

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Declaration

We hereby declare that this report titled “Hotel Management System” has been prepared as part of the CS545

– Software Engineering course in the MCA programme at the Indian Institute of Information Technology

Vadodara. The report organizes the requirements, design, process-model analysis, modular design and test-case documentation provided for this academic project.

External references and resources used in preparing the report have been acknowledged. Any test results described as examples are expected results only; tests that have not actually been executed are marked “Not run.” Faculty certification and signatures have not been filled in by the authors of this document.

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Note: Page numbers in this contents table are approximate and may shift if the report is edited. The exact instructions for

Experiments 3–7 were not present in the supplied materials; those sections are not fabricated.

• Project Introduction

1.1 Background

A Hotel Management System (HMS) is a software system intended to organize key hotel operations in one place. The system described in this report covers room management, guest registration, room search and availability, reservations, check-in and check-out, service recording, billing, payments and operational reports.

1.2 Problem Statement

Manual records and disconnected workflows can make it difficult to track room availability, reservations, guest

details and payments consistently. A centralized system is proposed to reduce manual effort, avoid overlapping bookings, maintain guest and transaction records, and generate booking and revenue summaries.

1.3 Objectives

- Centralize room, guest, reservation, stay, service and billing information.
- Track room availability and prevent overlapping reservations.
- Support check-in/check-out and update room status.
- Generate bills and record payments.
- Provide occupancy, reservation and revenue reports.

1.4 Scope and Limitations

The documented scope includes login/logout, room and room-type management, guest registration, availability search, reservation creation/modification/cancellation, check-in/check-out, bill generation, payment recording, room-status updates and reports. No source code or actual application screenshots were supplied with the source materials, so this report documents the design and expected tests rather than claiming a working implementation.

1.5 Report Organization

The report presents the SRS, process-model comparison, modular design and test-case design, followed by architecture/workflow diagrams, project planning, risk management, conclusion and references.

- # Experiment 1 – Software Requirements

SPECIFICATION (SRS)

2.1 Aim

To prepare a Software Requirements Specification (SRS) for a Hotel Management System by identifying its purpose, scope, stakeholders, functional and non-functional requirements, business rules and external interfaces.

2.2 Purpose

The purpose of the HMS is to provide a centralized and organized way to manage hotel operations and improve the consistency of booking, guest, stay and billing records.

2.3 Problem Statement

- Manual record keeping consumes staff time.
- Room availability and reservations must be tracked accurately.
- Double booking must be prevented.
- Guest details and payment records must be maintained.
- Booking and revenue reports must be generated.

2.4 Scope

Included area Scope

Access control Login and logout for authorized users.

Room management Room/type details, rates and room status.

Guest management Guest registration and updates.

Reservations Search availability; create, modify and cancel bookings.

Stay management Check-in, room assignment and checkout.

Billing and payments Generate invoices, calculate charges/tax and record payments. Reporting Booking summaries, occupancy and revenue reports.



Schema Of Hotel Management System

2.5 Stakeholders / User Classes

User class Responsibilities

Administrator Manage users, rooms, reports and system settings.

Receptionist / Hotel Staff Handle bookings, guest registration, check-in, checkout and billing.

Guest / Customer Make reservations and use hotel services. Hotel

Manager Monitor occupancy and revenue.

2.6 Assumptions

- Each room has a unique room number.
- Each room has a defined type, rate and status.
- Guests provide valid personal and contact information.
- Administrative and transaction functions are available only to authorized staff.
- A reliable database stores reservations, guests, rooms, bills and payments.

3 Experiment 1 – Srs (Continued)

3.1 Aim

To prepare a Software Requirements Specification (SRS) for a Hotel Management System by identifying its purpose, scope, stakeholders, functional and non-functional requirements, business rules, and external interfaces.

3.2 Problem Statement

The Hotel Management System is designed to computerize and simplify hotel operations such as room management, guest registration, room booking, check-in, check-out, billing, payment recording, and report generation. The system will help hotel staff maintain accurate records, reduce manual work, improve room utilization, and provide efficient service to guests.

3.3 Software Requirements Specification (SRS)

Purpose

The purpose of the proposed Hotel Management System is to provide a centralized system for managing hotel rooms, guests, reservations, check-in/check-out activities, billing, payments, and

hotel records. It will help receptionists and administrators perform daily operations quickly and accurately while providing guests with convenient booking and reservation services.

Scope

The system will cover the major operational activities of a hotel. Its scope includes user login, room and room-type management, guest registration, room search and availability checking, reservation/booking, check-in, check-out, bill generation, payment recording, reservation cancellation, and basic management reports. Authorized staff will be able to maintain hotel records and monitor room status.

Stakeholders / User Classes

S. No. **Stakeholder/User Class**

1. Administrator (Manages users, rooms, reports, and system settings)
2. Receptionist/Hotel Staff (Handles bookings, check-in, check-out, and billing)
3. Guest/Customer (Makes reservations and uses hotel services)
4. Hotel Manager (Monitors operations, occupancy, and revnue reports)

Assumptions

Each room has a unique room number and a defined room type, rate, and status.

Guests provide valid personal and contact information during registration or booking.

Only authorized hotel staff can access administrative and transaction functions.

The system has access to a reliable database for storing reservations, guests, rooms, bills, and payment records.

3.4 Functional Requirements

ID	**Requirement**
FR-01	User can log in and log out of the system.
FR-02	Receptionist can add new customer deatils.
FR-03	System can check room availability.
FR-04	System can book rooms for customers.
FR-05	System can update or cancel reservations.
FR-06	System can generate customer bills.
FR-07	System can manage room details and status.
FR-08	Admin can generate reports of bookings and revenue.

3.5 Non-Functional Requirements

ID	**Requirement**
NFR-01	System should be available 24x7.
NFR-02	Response time should be less than 3 seconds.
NFR-03	Customer data should be secure and password protected. NFR-04 System should support multiple users simultaneously.

3.6 Constraints / Business Rules

ID	**Requirement**
BR-01	A room cannot be booked if it is already occupied.
BR-02	Every booking must have a unique booking ID.
BR-03	Customer must provide valid identification before check-in. BR-04 Payment must be completed before check-out.

3.7 External Interface Requirements

User Interface

Major screens/interfaces include Login, Dashboard, Room Management, Room Availability/Search, Guest Registration, Reservation/Booking, Reservation Cancellation, Check-in, Check-out, Billing, Payment, Guest/Reservation Details, and Reports.

Input Requirements

S. No.	**Input Requirement**
1	Guest details (Name, Phone Number, Email, Address, ID Proof)
2	Room booking details (Room Type, Check-in Date, Check-out Date, Number of Guests.3 Payment details (Payment Method, Amount, Transaction ID) 4 Login credentials (Username and Password).

Output Requirements

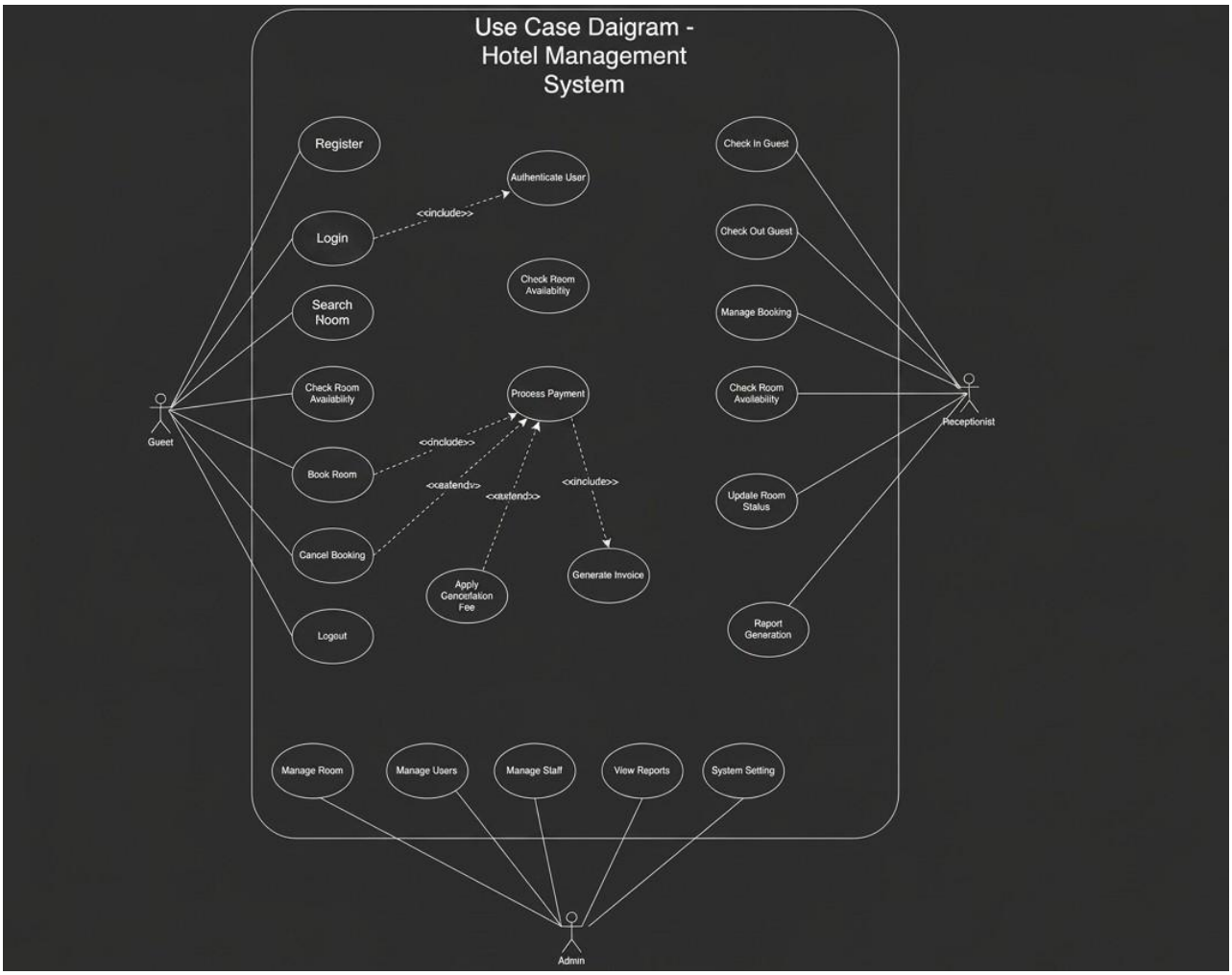
S. No.	**Output Requirement**
1	Booking confirmation with Booking ID
2	Customer bill/invoice and payment receipt
3	Reports (Room Occupancy, Reservation Summary, Revenue Reports)

3.8 Requirement Review

1. Are all requirements clear and understandable? Yes. The requirements are written as specific statements describing the expected behavior of the Hotel Management System.
2. Are any requirements contradictory? No. The listed requirements are consistent with the normal reservation, check-in, check-out, and billing workflow of a hotel.
3. Are the requirements testable? Yes. For example, FR-03 can be tested by entering a date range and room type and verifying that only rooms available for the requested period are displayed.

3.9 Result

In this experiment, a Software Requirements Specification for the Hotel Management System was prepared. The system scope, stakeholders, assumptions, functional and non-functional requirements, business rules, and interfaces were identified and documented.



4 Data Flow Diagram

Project Details

S. No.	Particulars	Description
1	Project Name	Hotel Management System
2	Project Objective	To manage hotel activities efficiently and reduce manual work.
3	Main Functions	Room booking, guest registration, room allocation, billing, payment, check-in and check-out.
4	Users	Hotel Admin, Receptionist and Guest

5	Benefits	Saves time, reduces errors and maintains accurate records.

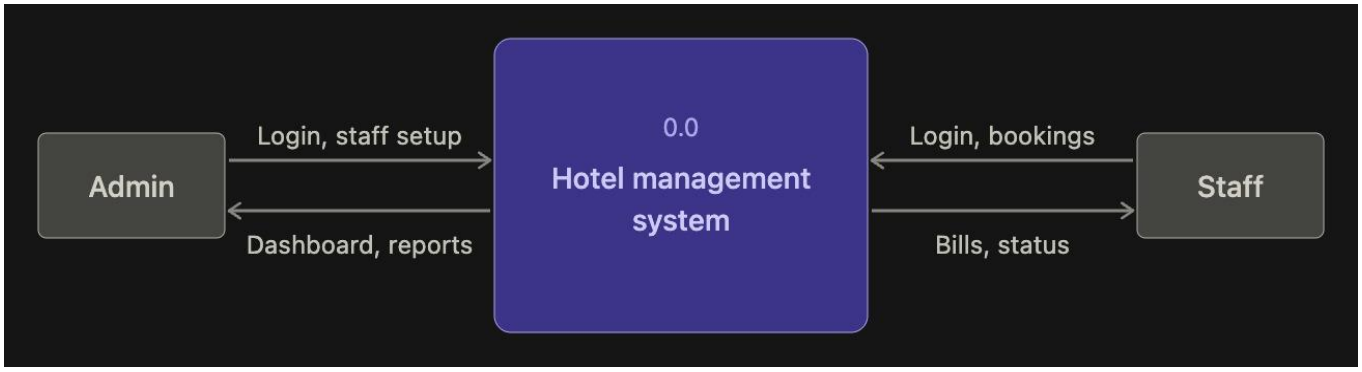
2. Major Processes of Hotel Management System

Process ID	Process Name	Description
P1	Room Reservation	Checks room availability and manages room bookings.
P2	Guest Registration	Records guest details such as name, address and contact number.
P3	Room Allocation	Assigns available rooms to registered guests.
P4	Billing and Payment	Calculates bills and records payment details.
P5	Check-in and Check-out	Manages guest arrival and departure.
P6	Report Generation	Generates reports related to bookings, rooms and payments.

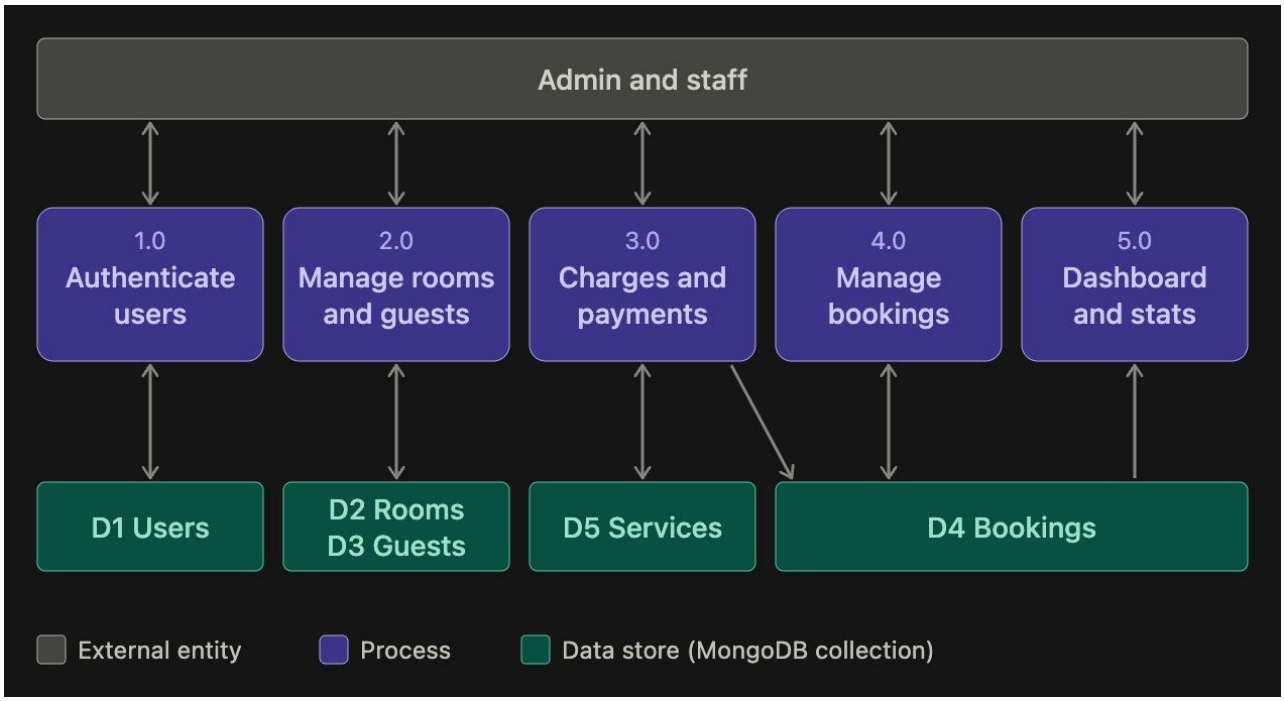
3. Data Stores of Hotel Management System

Data Store ID	Data Store Name	Information Stored
D1	Guest Database	Guest name, address, contact number and ID details.
D2	Room Database	Room number, room type, price and availability status.
D3	Reservation Database	Booking ID, reservation date, check-in and check-out dates.
D4	Payment Database	Bill amount, payment method and transaction details.
D5	Service Database	Food, laundry and other additional hotel services.

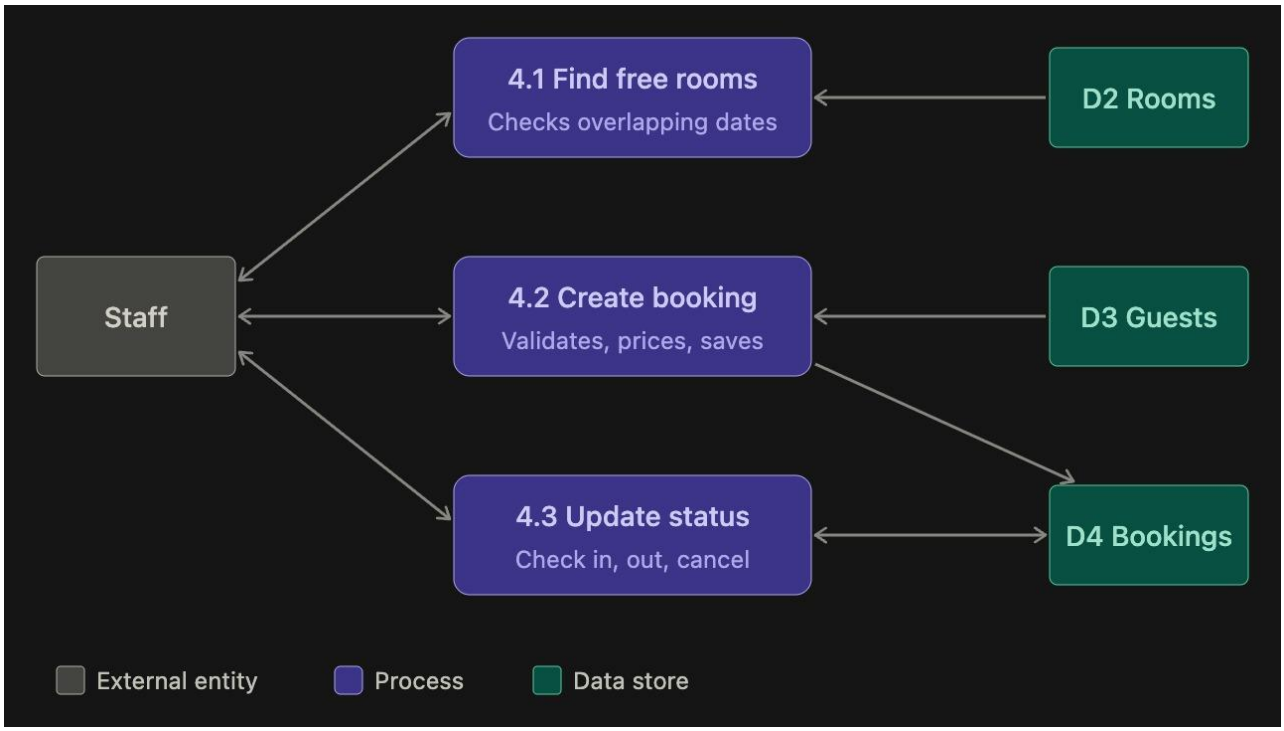
LEVEL 0 DFD



LEVEL 1 DFD



LEVEL 2 DFD



5 UML Diagram Theory – Hotel Management System

5.1

S. No.	Topic	Description
1	UML	Unified Modeling Language
2	Purpose	Represents the structure and behavior of the Hotel Management System.
3	Main Elements	Classes, attributes, methods, and relationships.
4	Objective	To manage guests, rooms, reservations, and payments efficiently.

5.2. Identification of Candidate Classes

S. No.	Candidate Class	Description
1	Guest	Stores guest information.
2	Room	Manages room details and availability.
3	Reservation	Handles room bookings.
4	Payment	Manages guest payments.

5	**Receptionist**	**Handles guest registration and room allocation.**
6	**Hotel**	**Maintains hotel information and room records.**
7	**Service**	**Manages additional hotel services.**
8	**Bill**	**Generates and maintains guest bills.**

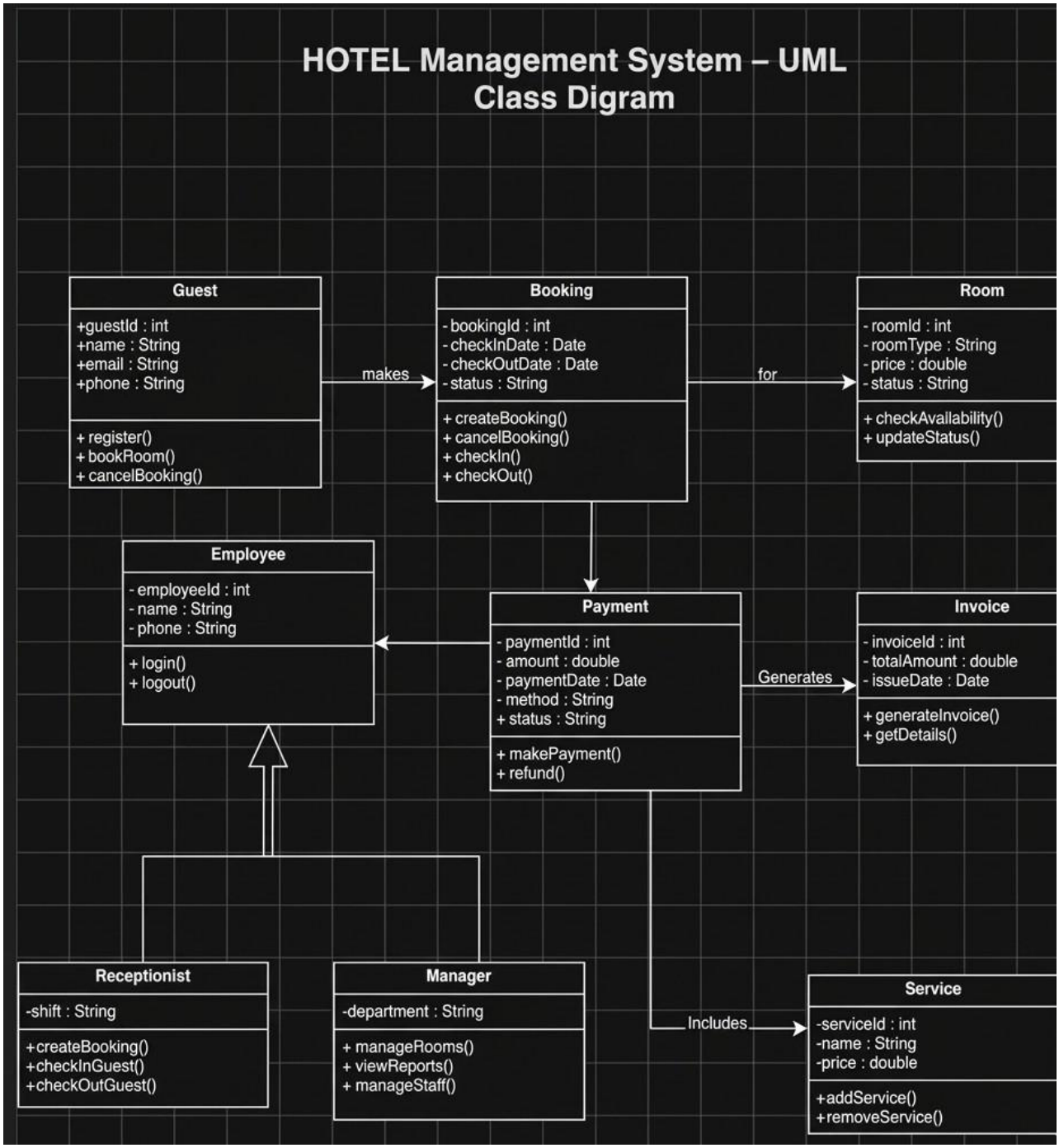
5.3. Attributes and Methods of Classes

Class Name	Attributes	Methods
Guest	**guestId, name, address, phone**	**registerGuest(), updateDetails()**
Room	**roomId, roomNumber, roomType, price, status**	**checkAvailability(), updateStatus()**
Reservation	**reservationId, checkInDate, checkOutDate, bookingStatus**	**bookRoom(), cancelReservation()**
Payment	**paymentId, amount, paymentDate, paymentMode**	**makePayment(), generateReceipt()**
Receptionist	**receptionistId, name, phone**	**registerGuest(), allocateRoom()**
Hotel	**hotelId, hotelName, address**	**addRoom(), viewRooms()**
Service	**serviceId, serviceName, serviceCharge**	**addService(), calculateCharge()**
Bill	**billId, totalAmount, billDate**	**calculateBill(), generateBill()**

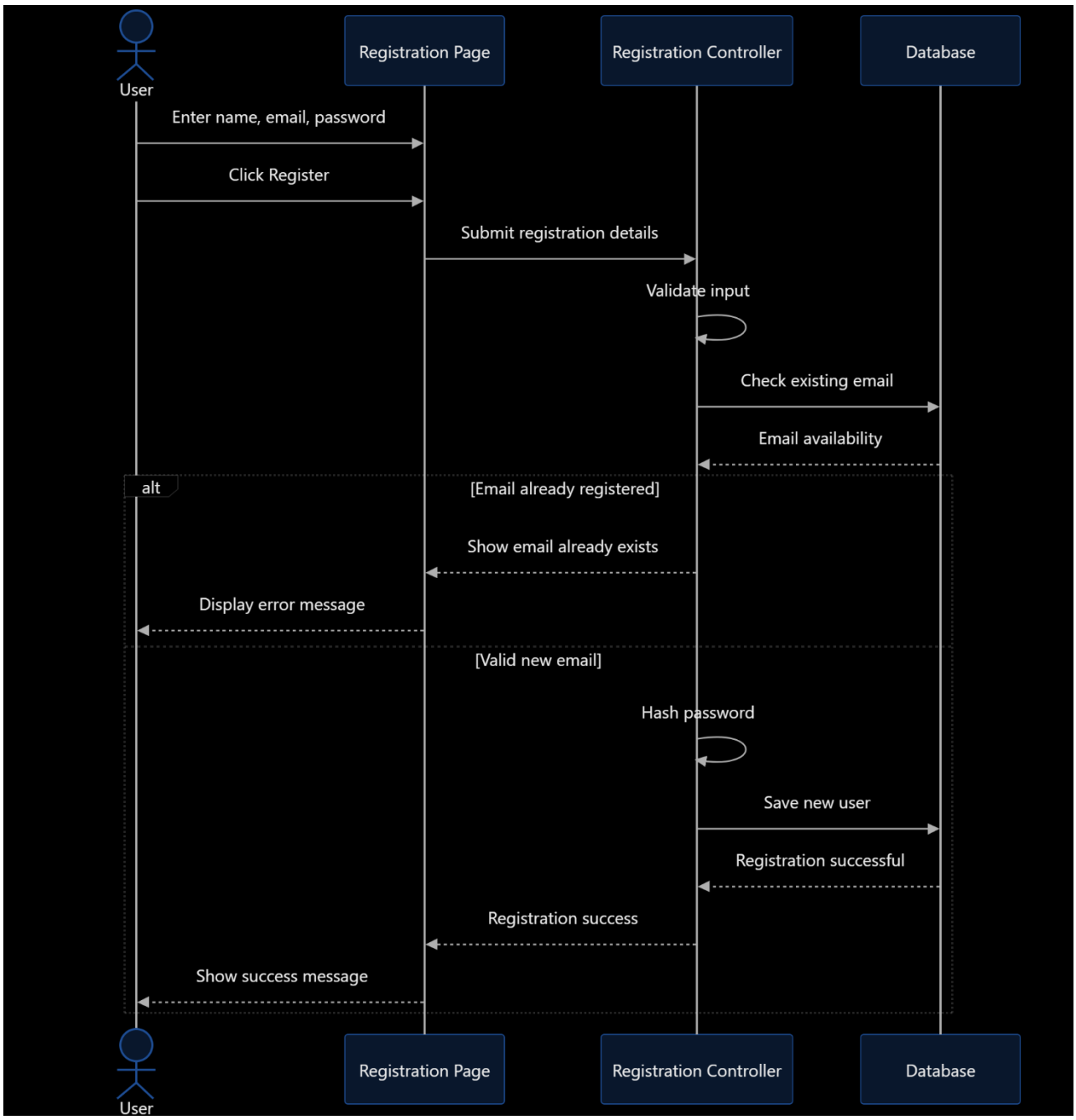
5.4 Relationship and Multiplicity – Hotel Management System

S. No.	Class 1	Relationship	Class 2	Multiplicity
1	Guest	Makes	Reservation	1 : Many (1:N)
2	Reservation	Allocates	Room	Many : 1 (N:1)
3	Guest	Makes	Payment	1 : Many (1:N)
4	Receptionist	Manages	Reservation	1 : Many (1:N)

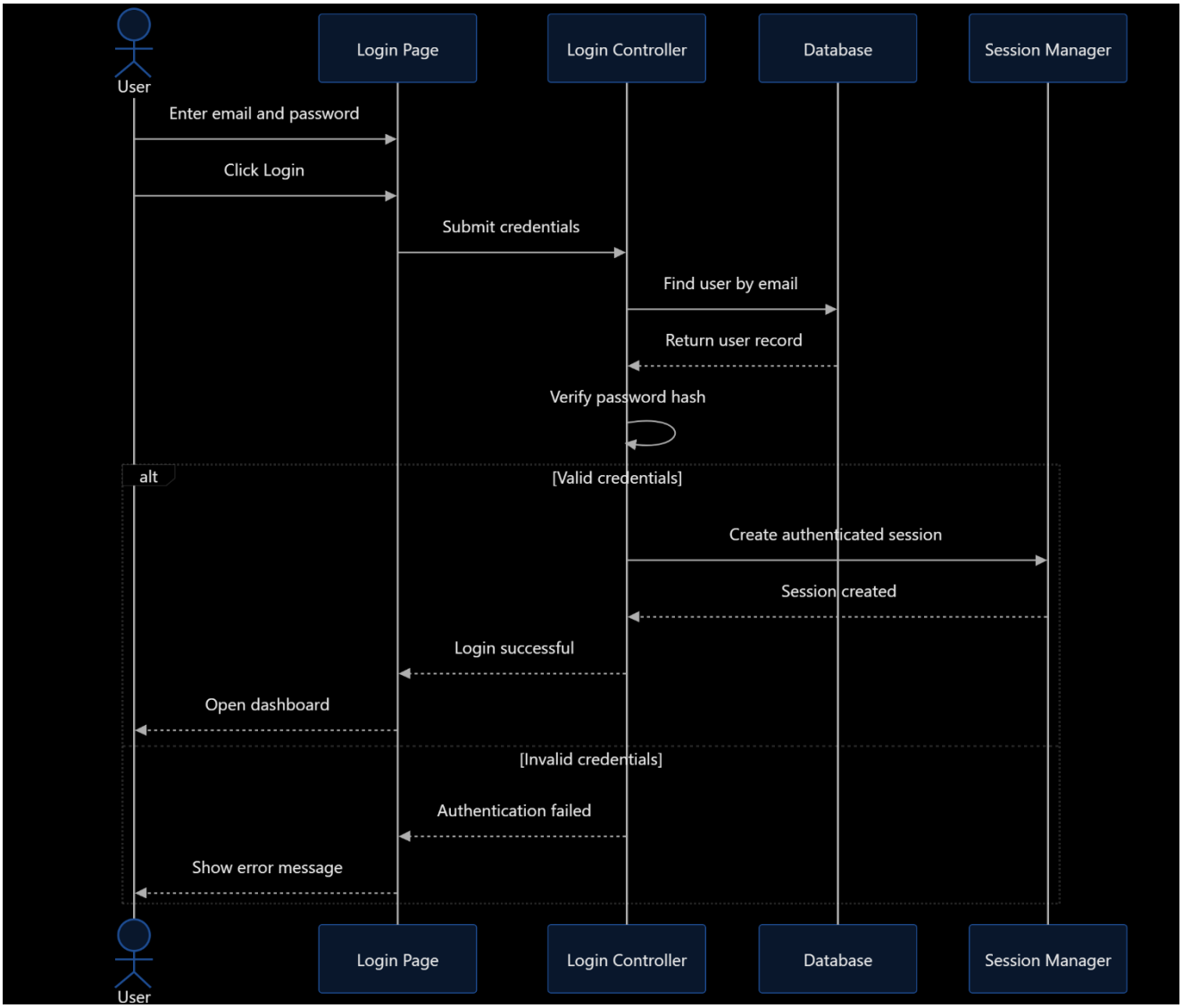
5	Hotel	Contains	Room	1 : Many (1:N)
6	Guest	Uses	Service	Many : Many (M:N)
7	Reservation	Generates	Bill	1 : 1
8	Bill	Records	Payment	1 : Many (1:N)



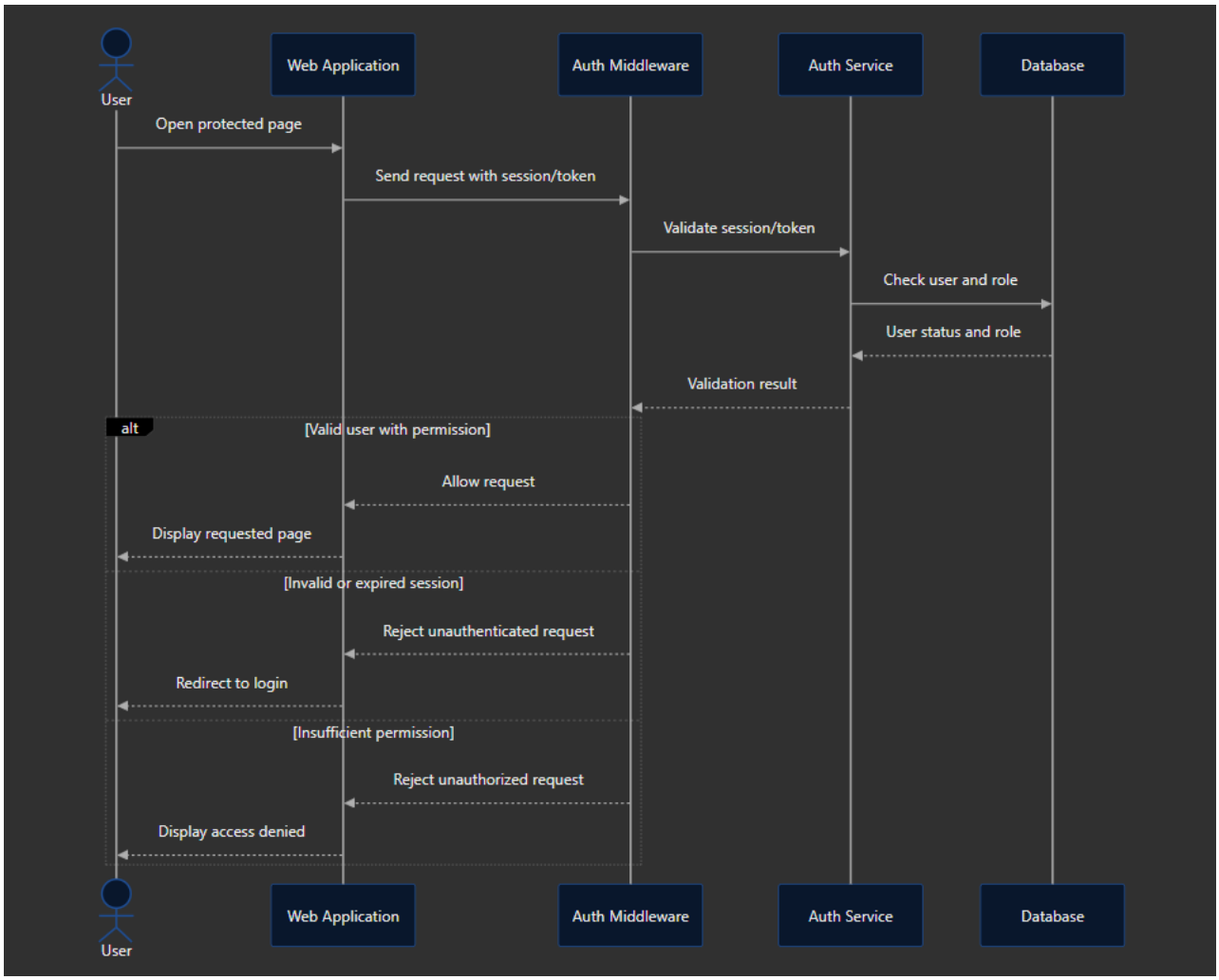
6 User Registration Sequence Diagram



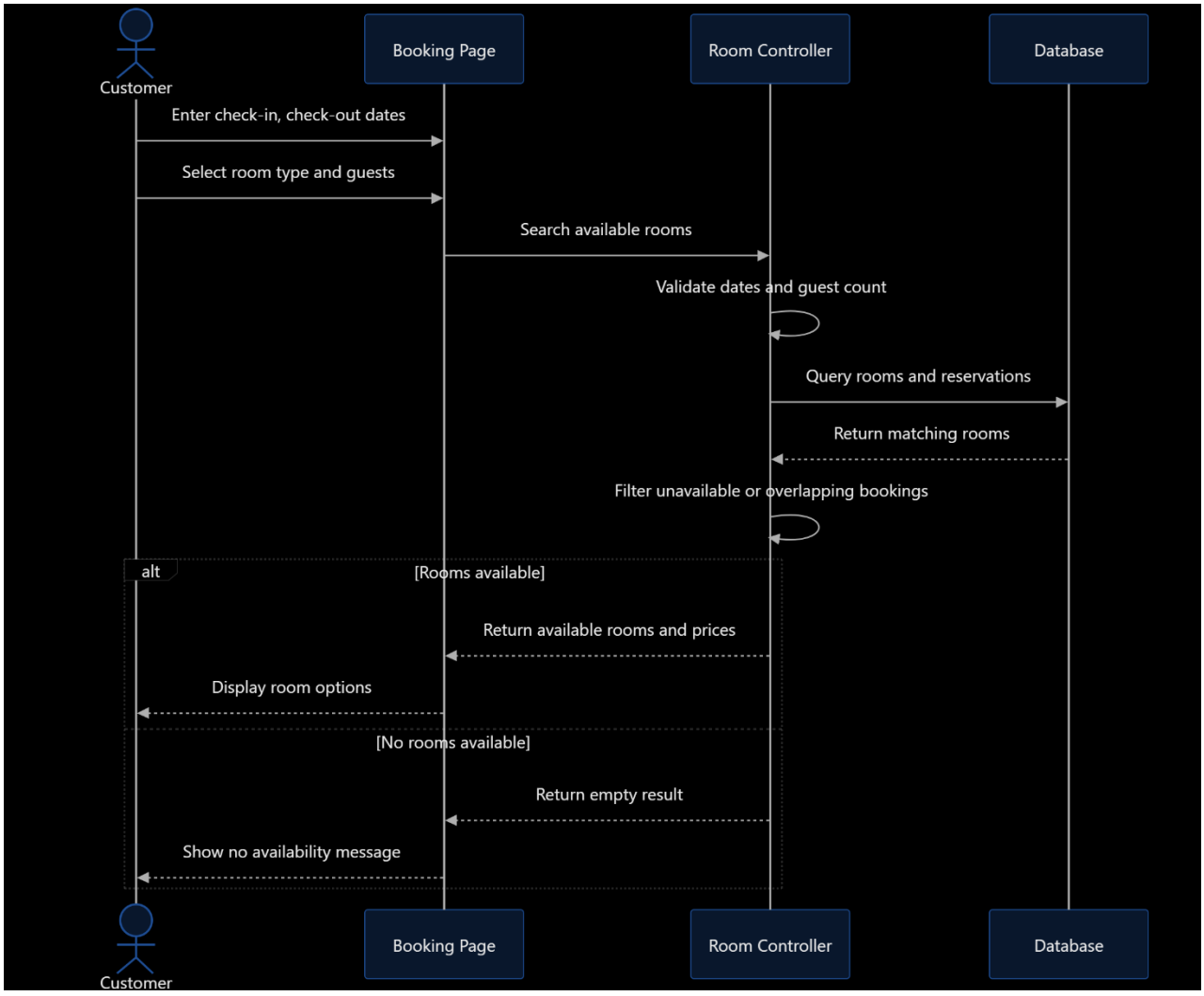
User Login Sequence Diagram



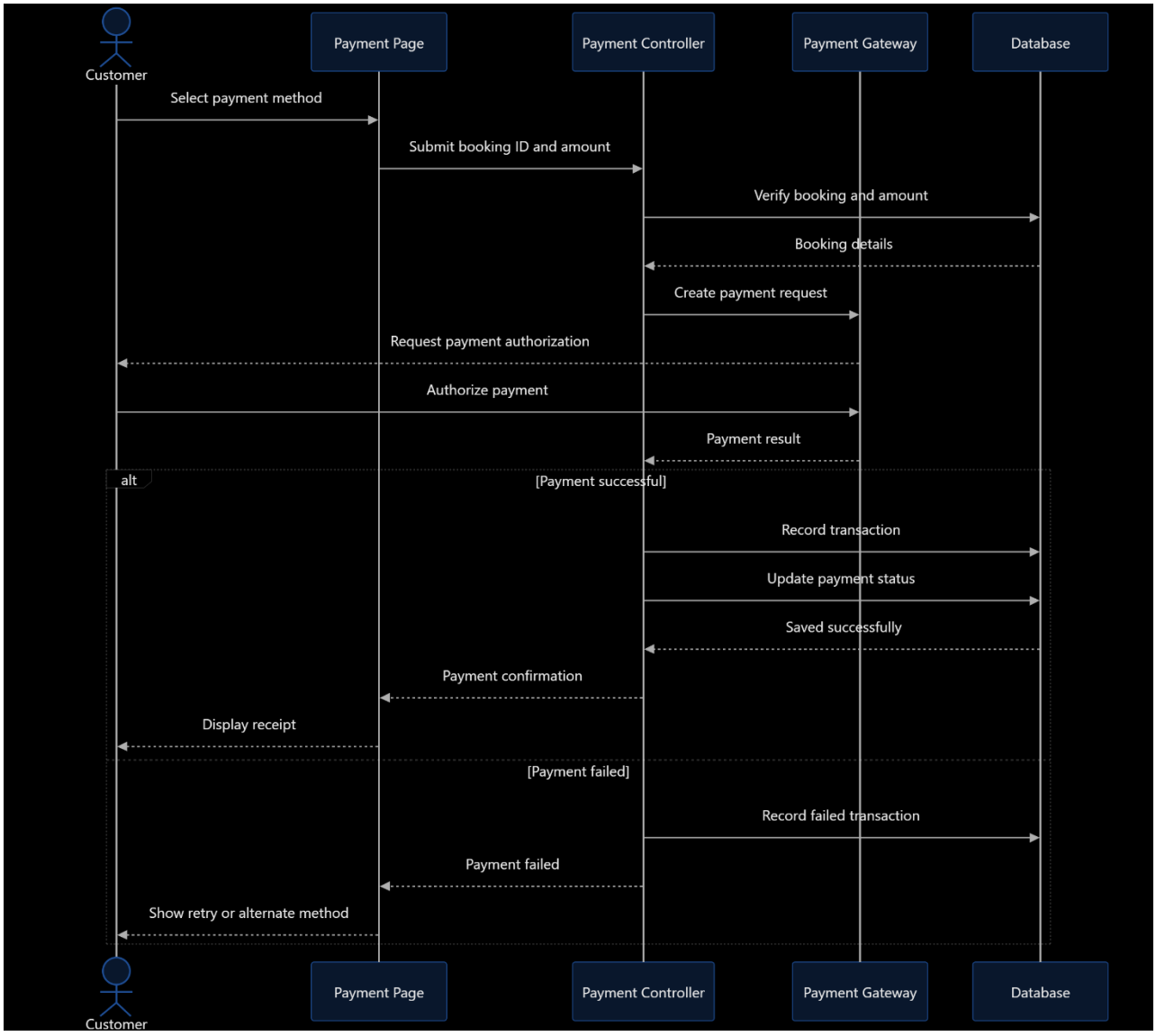
Authentication & Authorization Sequence Diagram



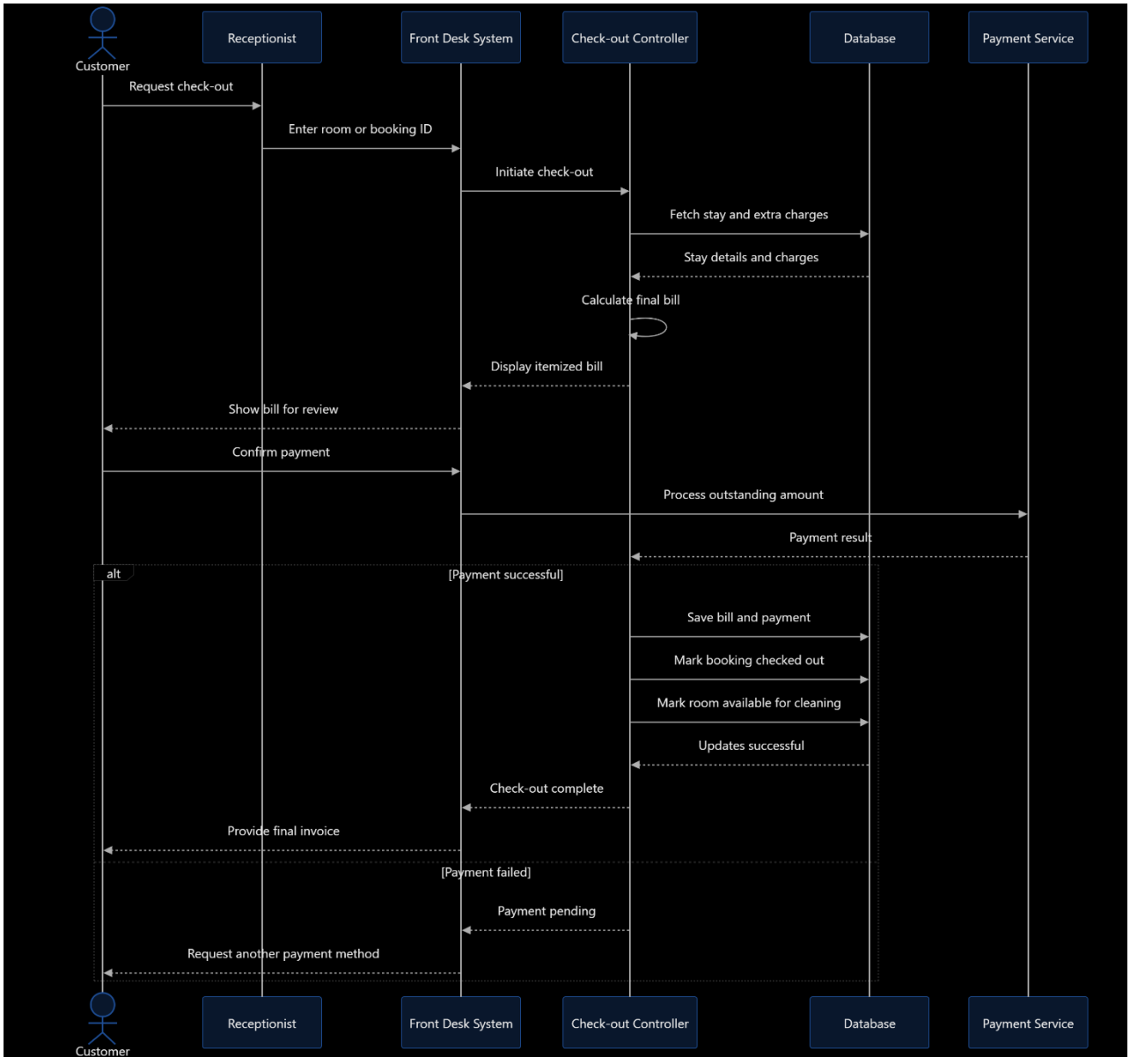
Room Availability Search Sequence Diagram



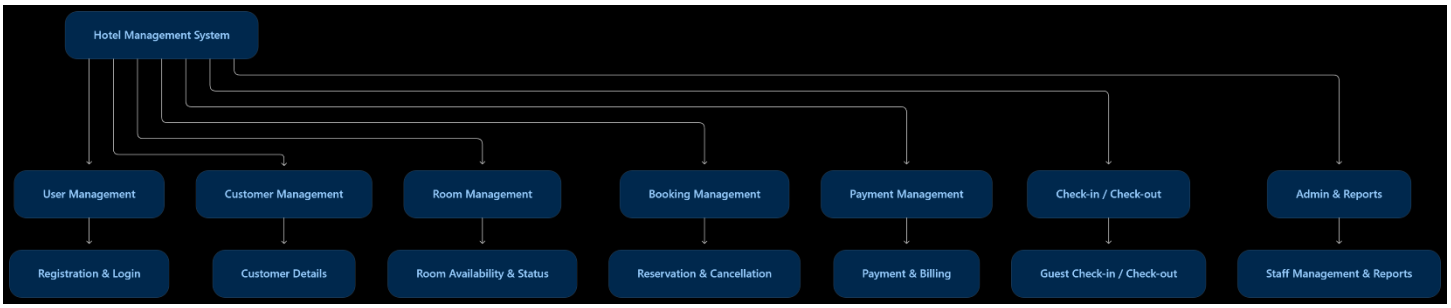
Payment Processing Sequence Diagram



Check-out & Bill Generation Sequence Diagram



Module Diagram



Functional Requirements

Sr. No.	Module	Functional Requirements
1	User Management	The system shall allow users to register, log in, and log out.
2	Authentication	The system shall verify user identity and access permissions.
3	Customer Management	The system shall allow customer details to be added, viewed, and updated.
4	Room Management	The system shall allow rooms, prices, and room availability to be managed.
5	Booking Management	The system shall allow room reservations, bookings, and cancellations.
6	Payment Management	The system shall process payments and generate bills and receipts.
7	Check-in / Check-out	The system shall record guest check-in and check-out details.
8	Staff Management	The system shall manage staff information and roles.

9	Report Management	The system shall generate booking, revenue, and room status reports.
10	Admin Management	The administrator shall manage and monitor all system modules.

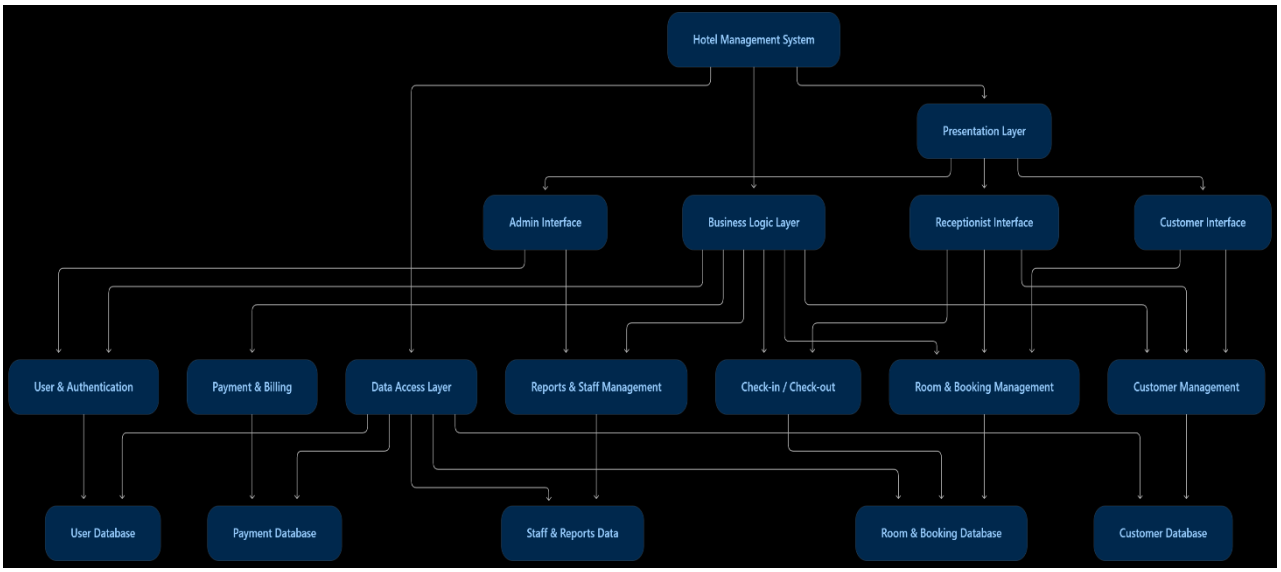
Module Identification

Module ID	Module Name	Description	Main User
M1	User Management	Handles user registration, login, and account management.	Customer/ Admin
M2	Authentication	Verifies user credentials and controls system access.	All Users
M3	Customer Management	Manages customer personal information and profiles.	Receptionist/ Admin
M4	Room Management	Manages room details, prices, and availability.	Admin/ Receptionist
M5	Booking Management	Handles reservations, bookings, and cancellations.	Customer/ Receptionist
M6	Payment Management	Manages payments, bills, receipts, and refunds.	Customer/ Cashier
M7	Check-in / Check-out	Records guest arrivals and departures.	Receptionist
M8	Staff Management	Manages staff accounts, details, and roles.	Admin
M9	Report Management	Generates booking, revenue, and occupancy reports.	Admin
M10	Admin Dashboard	Provides centralized system monitoring and control.	Admin

4. Interface Between Modules

Sr. No.	Module 1	Module 2	Interface / Data Exchange
1	User Management	Authentication	User ID, email, and login credentials.
2	Authentication	Admin Dashboard	User role and access permissions.
3	Customer Management	Booking Management	Customer ID and customer details.
4	Room Management	Booking Management	Room ID, room price, and availability status.
5	Booking Management	Payment Management	Booking ID and total amount.
6	Payment Management	Booking Management	Payment status and transaction ID.
7	Booking Management	Check-in / Check-out	Reservation details and booking status.
8	Check-in / Check-out	Room Management	Room occupancy status.
9	Booking Management	Report Management	Booking and cancellation data.
10	Payment Management	Report Management	Payment and revenue data.
11	Staff Management	Authentication	Staff account details and assigned roles.
12	Admin Dashboard	All Modules	Administrative controls and module management.

Modular Diagram



How Did We Reduce the Dependency? — Hotel Management System

Sr. No.	Technique	How Dependency Is Reduced
1	Modularization	The system is divided into independent modules such as Customer, Room, Booking, and Payment Management.
2	Loose Coupling	Modules communicate through defined interfaces instead of directly depending on each other's internal implementation.
3	High Cohesion	Each module performs a specific task, making it easier to maintain and modify.
4	Interface-Based Communication	Modules exchange data through predefined interfaces and service methods.
5	Database Abstraction	A data access layer separates business logic from database operations.

6	Separation of Concerns	User interface, business logic, and database operations are maintained separately.
7	Reusable Components	Common functions, such as authentication and billing, are reused instead of duplicated.
8	Dependency Injection	Required services are provided to modules rather than created directly inside them.

- **Hotel Management System — Functional Testing Documentation**

1. Functional Test Case Design

Test Case ID	Module	Requirement Tested	Test Scenario	Expected Result
TC-01	User Registration	FR-01	Register with valid details	Account is created successfully
TC-02	User Registration	FR-01	Register with an existing email	Duplicate registration is rejected
TC-03	Login	FR-02	Login with valid credentials	User is redirected to the dashboard
TC-04	Login	FR-02	Login with an incorrect password	Error message is displayed
TC-05	Authentication	FR-03	Access a protected page without login	Access is denied or user is redirected to login
TC-06	Customer Management	FR-04	Add a new customer	Customer record is saved
TC-07	Customer Management	FR-04	Update customer information	Updated details are saved
TC-08	Room Management	FR-05	Add a new room	Room is added successfully
TC-09	Room Management	FR-05	Search available rooms for selected dates	Only available rooms are displayed
TC-10	Booking Management	FR-06	Book an available room	Reservation is created

TC-11	Booking Management	FR-06	Book an already reserved room for overlapping dates	Booking is rejected
TC-12	Booking Management	FR-07	Cancel an eligible reservation	Booking status changes to cancelled
TC-13	Payment Management	FR-08	Make a successful payment	Payment is recorded and confirmation is displayed
TC-14	Payment Management	FR-08	Simulate a failed payment	Failure is recorded and the user is notified
TC-15	Check-in	FR-09	Check in with a valid reservation	Guest check-in is recorded
TC-16	Check-out	FR-09	Check out after a valid stay	Final bill is generated and checkout is recorded
TC-17	Staff Management	FR-10	Admin adds a staff member	Staff record is saved
TC-18	Reports	FR-11	Generate a revenue report	Report displays the relevant payment data
TC-19	Authorization	FR-03	Customer attempts an admin-only operation	Access is denied
TC-20	Billing	FR-08	Calculate the bill for room charges and extra services	Total amount is calculated correctly

2. Requirements Tested

Requirement ID	Functional Requirement	Description	Priority
FR-01	User Registration	The system shall register new users and prevent duplicate accounts.	High
FR-02	User Login	The system shall authenticate users using valid credentials.	High
FR-03	Authentication and Authorization	The system shall restrict access according to user roles.	High
FR-04	Customer Management	The system shall add, view, and update customer details.	High

FR-05	Room Management	The system shall manage room information, prices, and availability.	High
FR-06	Booking Management	The system shall create reservations and prevent overlapping bookings.	High
FR-07	Booking Cancellation	The system shall cancel eligible reservations and update their status.	Medium
FR-08	Payment and Billing	The system shall process payments and generate accurate bills.	High
FR-09	Check-in and Check-out	The system shall record guest arrival and departure.	High
FR-10	Staff Management	The system shall allow authorized administrators to manage staff.	Medium
FR-11	Report Management	The system shall generate booking, occupancy, and revenue reports.	Medium

3. Functional Requirements Across Services

Service / Module	Input	Processing	Expected Output	Dependency
Registration Service	Name, email, password	Validate fields and check duplicate email	Account creation status	User Database
Authentication Service	Email, password, user role	Verify credentials and permissions	Login status and authorized access	User Database
Customer Service	Customer ID and profile details	Validate and store customer information	Customer record	Customer Database
Room Service	Room ID, type, price, status	Add, update, and search rooms	Room details and availability	Room Database
Booking Service	Customer ID, room ID, dates	Check availability and create reservation	Booking ID and booking status	Customer and Room Services
Cancellation Service	Booking ID, cancellation request	Verify booking status and cancellation rules	Cancellation confirmation	Booking Service

Payment Service	Booking ID, amount, payment method	Validate amount and process transaction	Payment status and transaction reference	Booking and Payment Database
Billing Service	Room charges, extra services, taxes	Calculate final payable amount	Itemized bill or invoice	Booking and Payment Services
Check-in Service	Booking ID and guest verification	Validate reservation and record arrival	Check-in confirmation	Booking and Room Services
Check-out Service	Booking ID and outstanding charges	Calculate final bill and update stay status	Checkout confirmation	Booking, Billing, and Room Services
Staff Service	Staff details and assigned role	Validate and save staff record	Staff creation or update status	User and Staff Database
Reporting Service	Date range and report type	Aggregate booking, payment, and occupancy data	Report output	Booking and Payment Services

4. Test Data and Assumptions

Sr. No.	Area	Test Data / Assumption
1	User Registration	Name: Rahul Sharma; Email: rahul.test@example.com; Password: Test@12345
2	Existing User	Email: existing.test@example.com is already registered
3	Valid Login	Registered email and correct password
4	Invalid Login	Registered email and incorrect password
5	Customer Details	Customer ID: C101; Phone: 9876543210
6	Room Details	Room ID: R101; Type: Deluxe; Rate: ₹3,000 per night
7	Available Room	Room R101 is available for the selected dates
8	Booked Room	Room R102 has an existing reservation for the same dates
9	Booking Dates	Check-in: 10 November 2026; Check-out: 12 November 2026
10	Booking Duration	Two nights
11	Payment Success	Valid test payment gateway response

12	Payment Failure	Simulated declined transaction
13	Check-in	Confirmed booking with valid guest details
14	Check-out	Completed two-night stay with no outstanding unpaid charges
15	Staff Management	Staff ID: S101; Role: Receptionist
16	Authorization	Customer account attempts to access an admin-only page
17	Revenue Report	Report period: 1 November 2026 to 30 November 2026
18	Billing Assumption	Room charges and any applicable taxes or extra service charges are included according to configured rules

Assumptions: These are sample test values. The system has a database, defined user roles, configurable room prices, and a payment gateway or payment simulator. Test transactions should use a sandbox environment, not real customer payments.

5. Test Data Set

Data ID	Field	Sample Value	Purpose
TD-01	Full Name	Rahul Sharma	Valid registration
TD-02	Email	rahul.test@example.com	Valid unique email
TD-03	Password	Test@12345	Valid password format
TD-04	Email	existing.test@example.com	Duplicate email validation
TD-05	Password	Wrong@123	Invalid login test
TD-06	Customer ID	C101	Customer lookup
TD-07	Room ID	R101	Available room booking
TD-08	Room ID	R102	Overlapping booking validation
TD-09	Room Type	Deluxe	Room search
TD-10	Room Rate	₹3,000 per night	Billing calculation
TD-11	Check-in Date	10-11-2026	Reservation start date
TD-12	Check-out Date	12-11-2026	Reservation end date
TD-13	Number of Nights	2	Bill calculation
TD-14	Booking ID	B1001	Reservation lookup
TD-15	Payment Amount	₹6,000 before taxes and extras	Payment processing
TD-16	Payment Result	Success	Successful transaction
TD-17	Payment Result	Declined	Failed transaction
TD-18	Staff ID	S101	Staff management

TD-19	User Role	Customer	Restricted-access testing
TD-20	User Role	Admin	Authorized administrative testing

6. Test Case Register

Test Case ID	Test Case Title	Preconditions	Test Steps	Expected Result	Priority
TC-01	Successful Registration	Email is not registered	Enter valid details and submit	Account is created	High
TC-02	Duplicate Registration	Email already exists	Register using the existing email	Duplicate account is rejected	High
TC-03	Successful Login	User account exists	Enter valid credentials and log in	Dashboard opens	High
TC-04	Invalid Login	User account exists	Enter incorrect password	Error message appears	High
TC-05	Protected Page Access	User is logged out	Open a protected URL	Access is denied or login page opens	High
TC-06	Add Customer	Authorized user is logged in	Enter customer details and save	Customer record is created	High
TC-07	Update Customer	Customer record exists	Change customer details and save	Updated details are displayed	Medium

TC-08	Add Room	Admin is logged in	Enter room details and save	Room is created	High
TC-09	Search Available Rooms	Valid dates are entered	Search rooms for the selected dates	Available rooms are displayed	High
TC-10	Successful Booking	Room is available	Select room, enter dates, confirm booking	Reservation is created	High
TC-11	Duplicate Room Booking	Room is already booked for overlapping dates	Attempt another reservation	Conflicting booking is rejected	High
TC-12	Cancel Booking	Reservation is eligible for cancellation	Submit cancellation request	Booking is marked cancelled	Medium
TC-13	Successful Payment	Valid pending booking exists	Complete sandbox payment	Payment is recorded successfully	High
TC-14	Failed Payment	Pending booking exists	Simulate declined payment	Failure is shown; booking is not falsely marked paid	High
TC-15	Guest Check-in	Confirmed reservation exists	Verify booking and record arrival	Check-in is recorded	High
TC-16	Guest Check-out	Guest is checked in	Calculate charges and confirm checkout	Final bill and checkout status are saved	High
TC-17	Add Staff	Admin is authorized	Enter staff details and save	Staff record is created	Medium
TC-18	Generate Revenue Report	Payment records exist	Select report dates and generate	Correct revenue data is displayed	Medium
TC-19	Customer Access Restriction	Customer is logged in	Attempt an admin-only operation	Access is denied	High

TC-20	Bill Calculation	Booking and applicable charges exist	Calculate total for two nights plus configured extras and taxes	Total matches the billing rules	High

6. Requirements Traceability Matrix (RTM)

Requirement ID	Requirement Description	Related Test Case IDs	Test Coverage
FR-01	User Registration	TC-01, TC-02	Covered
FR-02	User Login	TC-03, TC-04	Covered
FR-03	Authentication and Authorization	TC-05, TC-19	Covered
FR-04	Customer Management	TC-06, TC-07	Covered
FR-05	Room Management	TC-08, TC-09	Covered
FR-06	Booking Management	TC-10, TC-11	Covered
FR-07	Booking Cancellation	TC-12	Covered
FR-08	Payment and Billing	TC-13, TC-14, TC-20	Covered
FR-09	Check-in and Check-out	TC-15, TC-16	Covered
FR-10	Staff Management	TC-17	Covered
FR-11	Report Management	TC-18	Covered

7. Test Execution Register

Test Case ID	Actual Result	Status	Defect ID	Remarks
TC-01	To be recorded during testing	Not Run	—	Pending execution
TC-02	To be recorded during testing	Not Run	—	Pending execution
TC-03	To be recorded during testing	Not Run	—	Pending execution
TC-04	To be recorded during testing	Not Run	—	Pending execution
TC-05	To be recorded during testing	Not Run	—	Pending execution
TC-06	To be recorded during testing	Not Run	—	Pending execution
TC-07	To be recorded during testing	Not Run	—	Pending execution
TC-08	To be recorded during testing	Not Run	—	Pending execution

TC-09	To be recorded during testing	Not Run	—	Pending execution
TC-10	To be recorded during testing	Not Run	—	Pending execution
TC-11	To be recorded during testing	Not Run	—	Pending execution
TC-12	To be recorded during testing	Not Run	—	Pending execution
TC-13	To be recorded during testing	Not Run	—	Pending execution
TC-14	To be recorded during testing	Not Run	—	Pending execution
TC-15	To be recorded during testing	Not Run	—	Pending execution
TC-16	To be recorded during testing	Not Run	—	Pending execution
TC-17	To be recorded during testing	Not Run	—	Pending execution
TC-18	To be recorded during testing	Not Run	—	Pending execution
TC-19	To be recorded during testing	Not Run	—	Pending execution
TC-20	To be recorded during testing	Not Run	—	Pending execution

8. Test Environment and Assumptions

Sr. No.	Parameter	Description
1	Application	Hotel Management System
2	Frontend	Web browser or application interface
3	Backend	Application server and business logic
4	Database	Configured relational database
5	Browser	Chrome, Firefox, or another supported browser
6	Test Environment	Development or staging environment
7	Payment Environment	Sandbox payment gateway or simulator
8	Test Accounts	Customer, receptionist, and admin accounts
9	Test Data	Sample users, rooms, bookings, and payments
10	Security	Role-based access and password validation are enabled

9. Entry and Exit Criteria

Criteria Type	Criteria	Description
Entry Criteria	Requirements Available	Functional requirements are documented and reviewed.

Entry Criteria	Application Available	The application is deployed in the test environment.
Entry Criteria	Test Data Prepared	Required sample accounts, rooms, and bookings are available.
Entry Criteria	Test Cases Reviewed	Test cases and expected results are approved.
Exit Criteria	Test Execution Completed	All planned test cases have been executed.
Exit Criteria	Critical Defects Resolved	Critical and high-severity defects are fixed or formally accepted.
Exit Criteria	Requirements Verified	All required functions have been tested against their requirements.
Exit Criteria	Test Summary Prepared	Pass, fail, blocked, and not-run counts are documented.

10. Test Summary Report Template

Metric	**Value**
Total Test Cases	20
Passed	To be determined
Failed	To be determined
Blocked	To be determined
Not Run	20 initially
Defects Identified	To be determined
Overall Result	Pending test execution